

Multivariate Lab 3

Date: 14-02-2026

You are given the random vector $X' = [X_1, X_2, X_3, X_4]$ with mean vector $\mu'_x = [4, 3, 2, 1]$ and variance-covariance matrix

$$\Sigma_x = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}$$

Partition of X as $X = \begin{bmatrix} X_1 \\ X_2 \\ \dots \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} X^{(1)} \\ \dots \\ X^{(2)} \end{bmatrix}$

Let $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$ and consider the linear transformation $AX^{(1)}$ and $BX^{(2)}$.

Find (a) $E(X^{(1)})$, (b) $E(AX^{(1)})$ (c) $Cov(X^{(1)})$ (d) $Cov(AX^{(1)})$ (e) $E(X^{(2)})$ (f) $E(BX^{(2)})$ (g) $Cov(X^{(2)})$ (h) $Cov(BX^{(2)})$ (i) $Cov(X^{(1)}, X^{(2)})$ (j) $Cov(AX^{(1)}, BX^{(2)})$.

Solution:

Source: R Studio

Program:

```
A=c(1,2)
```

```
A
```

```
P=matrix(A, nrow=1, ncol=2)
```

```
P
```

```
B=c(1,2,-2,-1)
```

```
B
```

```
Q=matrix(B, nrow=2, ncol=2)
```

```
Q
```

```
Mean_vector=c(4,3,2,1)
```

```
Mean_vector
```

```
R=matrix(A1, nrow=1, ncol=4)
```

```
R
```

```
Cov_x=(c(3,0,2,2,0,1,1,0,2,1,9,-2,2,0,-2,4))
```

```
Cov_x
```

```
S=matrix(A2, nrow=4, ncol=4)
```

```
S
```

```
E_X1=Mean_vector[1:2]
```

```
E_X1
```

```
E_X2=Mean_vector[3:4]
```

```
E_X2
```

```
E_AX1=t(A)%*%E_X1
```

```
E_AX1
```

```
E_BX2=Q%*%E_X2
```

```

E_BX2
Cov_X1=S[1:2,1:2]
Cov_X1
Cov_X2=S[3:4,3:4]
Cov_X2
Cov_X1X2=S[1:2,3:4]
Cov_X1X2
Cov_AX1=P%%Cov_X1%%t(P)
Cov_AX1
Cov_BX2=Q%%Cov_X2%%t(Q)
Cov_BX2
Cov_AX1BX2=P%%Cov_X1X2%%Q
Cov_AX1BX2

```

OUTPUT:-

```

> A=c(1,2)
> A
[1] 1 2
> P=matrix(A, nrow=1, ncol=2)
> P
      [,1] [,2]
[1,]    1    2
> B=c(1,2,-2,-1)
> B
[1] 1 2 -2 -1
> Q=matrix(B, nrow=2, ncol=2)
> Q
      [,1] [,2]
[1,]    1   -2
[2,]    2   -1
> A1=c(Mean_vector)
> Mean_vector=c(4,3,2,1)
> Mean_vector
[1] 4 3 2 1
> R=matrix(A1, nrow=1, ncol=4)
> R
      [,1] [,2] [,3] [,4]
[1,]    4    3    2    1
> Cov_x=(c(3,0,2,2,0,1,1,0,2,1,9,-2,2,0,-2,4))
> Cov_x
[1] 3 0 2 2 0 1 1 0 2 1 9 -2 2 0 -2 4
> S=matrix(A2, nrow=4, ncol=4)
> S
      [,1] [,2] [,3] [,4]
[1,]    3    0    2    2
[2,]    0    1    1    0
[3,]    2    1    9   -2
[4,]    2    0   -2    4
> E_X1=Mean_vector[1:2]
> E_X1
[1] 4 3
> E_X2=Mean_vector[3:4]
> E_X2
[1] 2 1
> E_AX1=t(A)%%E_X1
> E_AX1

```

```

      [,1]
[1,]    10
> E_BX2=Q%%E_X2
> E_BX2
      [,1]
[1,]     0
[2,]     3
> Cov_X1=S[1:2,1:2]
> Cov_X1
      [,1] [,2]
[1,]     3     0
[2,]     0     1
> Cov_X2=S[3:4,3:4]
> Cov_X2
      [,1] [,2]
[1,]     9    -2
[2,]    -2     4
> Cov_X1X2=S[1:2,3:4]
> Cov_X1X2
      [,1] [,2]
[1,]     2     2
[2,]     1     0
> Cov_AX1=P%%Cov_X1%%t(P)
> Cov_AX1
      [,1]
[1,]     7
> Cov_BX2=Q%%Cov_X2%%t(Q)
> Cov_BX2
      [,1] [,2]
[1,]    33    36
[2,]    36    48
> Cov_AX1BX2=P%%Cov_X1X2%%Q
> Cov_AX1BX2
      [,1] [,2]
[1,]     8   -10

```

CONCLUSION:-

$$(a) E(X^{(1)}) = \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$$

$$(b) E(AX^{(1)}) = \begin{bmatrix} 1,1 \\ 1,1 \end{bmatrix} \begin{bmatrix} 10 \end{bmatrix}$$

$$(c) \text{Cov}(X^{(1)}) = \begin{bmatrix} 1,1 & 1,2 \\ 1,1 & 3 & 0 \\ 2,1 & 0 & 1 \end{bmatrix}$$

$$(d) \text{COV}(AX^{(1)}) = \begin{bmatrix} 1,1 \\ 1,1 \end{bmatrix} \begin{bmatrix} 7 \end{bmatrix}$$

$$(e) E(X^{(2)}) = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$(f) E(BX^{(2)}) = \begin{bmatrix} 1,1 \\ 1,1 \\ 2,1 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$(g) \text{Cov}(X^{(2)}) = \begin{bmatrix} [1] & [2] \\ [1,] & 9 & -2 \\ [2,] & -2 & 4 \end{bmatrix}$$

$$(h) \text{Cov}(BX^{(2)}) = \begin{bmatrix} [1] & [2] \\ [1,] & 33 & 36 \\ [2,] & 36 & 48 \end{bmatrix}$$

$$(i) \text{Cov}(X^{(1)}, X^{(2)}) = \begin{bmatrix} [1] & [2] \\ [1,] & 2 & 2 \\ [2,] & 1 & 0 \end{bmatrix}$$

$$(j) \text{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} [1] & [2] \\ [1,] & 8 & -10 \end{bmatrix}$$